core



File: core.kicad_sch

thermal



File: thermal.kicad_sch

LAYOUT STUDY ONLY: two real offboard NTCs, no charge/fabrication approval.

Sheet: /
File: power_integrated_study.kicad_sch

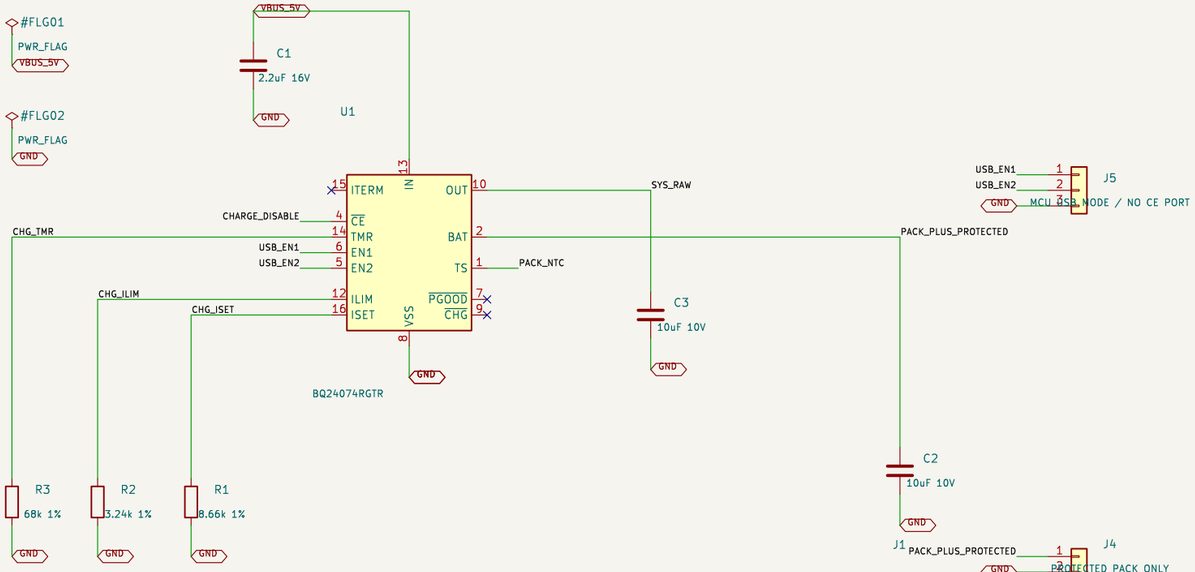
Title: Unified power safety candidate / NOFAB

Size: A4	Date:	Rev: WIDTH14-CANDIDATE
KiCad E.D.A. 9.0.2		Id: 1/3

2026-10-28 重建稿 / REBUILT — NO FAB — 禁止投板与连接未批准电池

不是原件恢复：本次重新生成与检查。JP1保持开路，充电默认禁用；已核IC landpattern；完整BOM、电池、NTC与热设计未放行。

1 电池电源路径 / POWER PATH

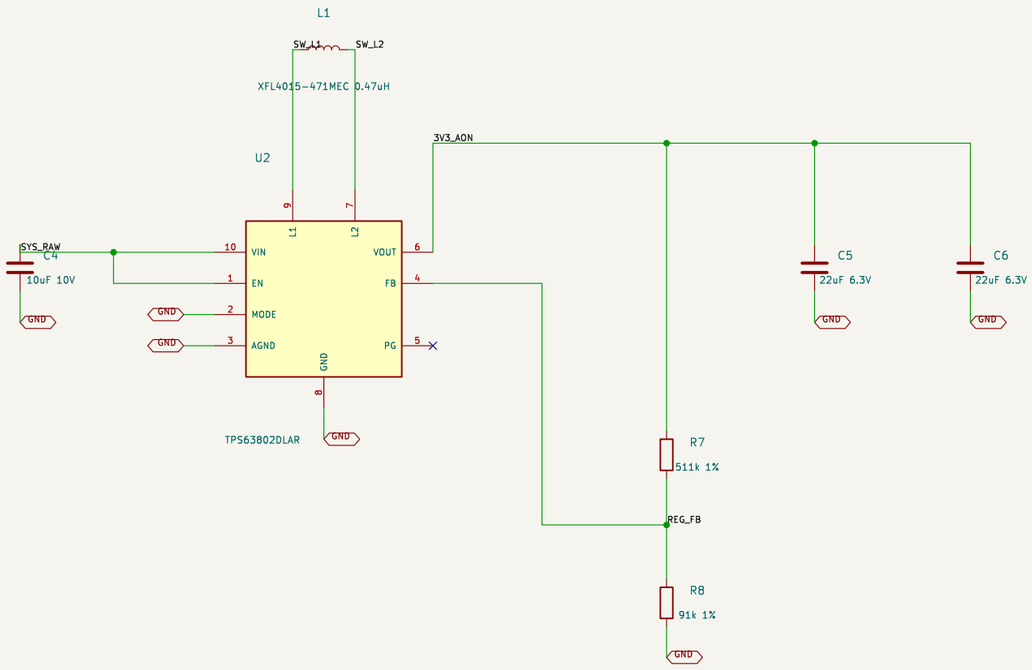


J1：外接真实NTC两线接口，型号/温度窗口/连接器待定。

GND = 保护后PACK_MINUS；禁止连接裸CELL_MINUS绕过保护。

RT102: OFFBOARD CELL NTC #1 via J1
Independent from RT101; protected GND only.
Candidate part, pack mounting NOT qualified.

2 常供电升降压 / BUCK-BOOST

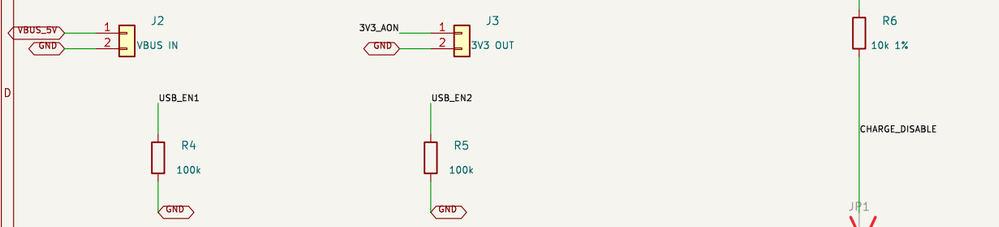


511k/91k -> 3.3077V标称；纹波/偏置/负载阶跃尚未测试。

COUT有效容量与电感饱和/DCR需按实际器件和布局核验。

14x45mm PCB候选；有效C、热、制造与线束尚未批准。

3 专用GPIO接口 / 调试充电跳线



USB_EN1 -> MCU GPIO18；USB_EN2 -> GPIO21；禁止复用UART。

复位默认LOW/LOW=USB100；休眠/枚举/来源授权仍需系统验证。

JP1为物理开路跳线，未批准电池/NTC前禁止短接。

R1=8.66k：91.12-113.73mA；R2=3.24k：最坏536.23mA。

R2不是500mA硬上限；CE支路电流也必须计入USB总电流。

RELEASE BLOCKERS / 未放行项目

仅验证过的5V输入；USB-C/ESD/OVP在外部。

VBUS_5V / 3V3_AON / USB_EN1 / USB_EN2为统一逻辑网名。

充电电压容差、NTC故障与温度边界、保护电池待验证。

CE内部约285k下拉；JP1开路也有VBUS偏置电流。

JP1改为外部安全许可串联开路；绝不是直接接地许可。

转换器低压能力不等于正常电池放电截止；策略待定。

本次ERC仅检查连接，不批准电池充电或生产。

SYSTEM-P0: core R6 and OPEN JP1 retained; NOFAB
非原件恢复；充电禁用；未批准生产

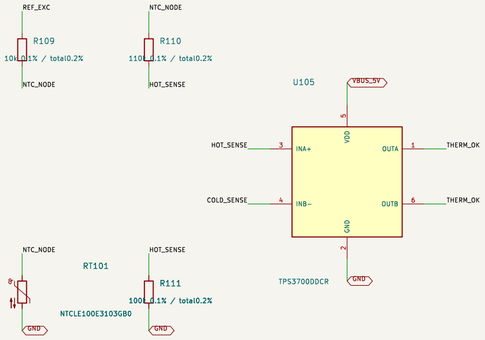
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File: core.kicad_sch

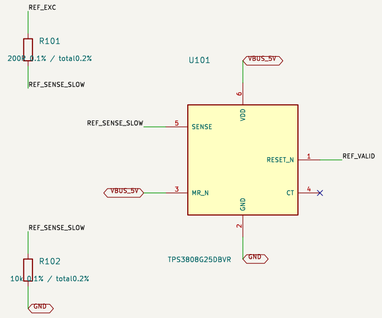
Title: 电源核心重建稿 / NO FAB

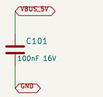
Size: A3	Date: 2026-10-28	**Rev: WIDTH14-CANDIDATE**
KiCad E.D.A. 9.0.2		Id: 2/3

CE dual-SPST logic interlock candidate / NEW-REF-P0 / NOFAB

SYSTEM-P0: thermal R8 / JP1 removed: CORE JP1 is sole OPEN commissioning gate.

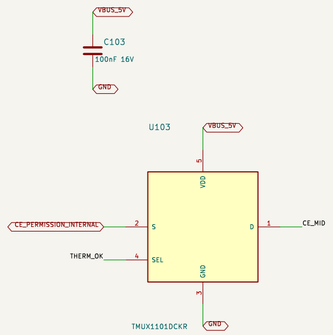


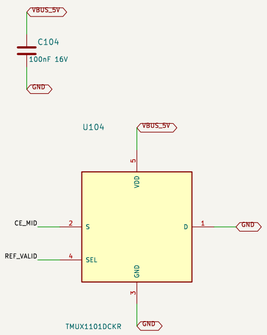




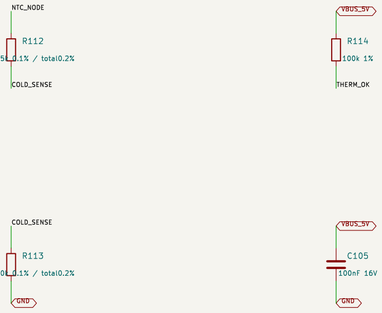


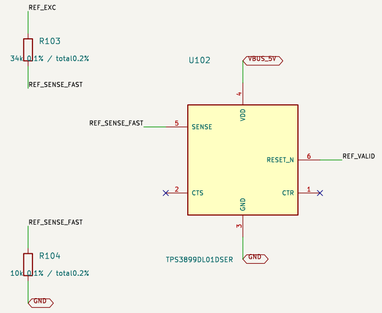


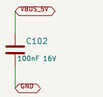












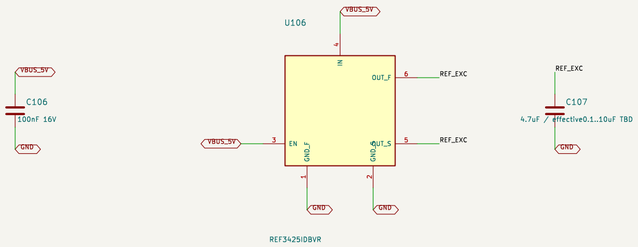
RT101 = OFFBOARD CELL NTC #2 / NOT PCB-MOUNTED
BOM retained: independent two wires via J104.
No pack qualification or thermal-bond approval.

U101 CT deliberately open: 12-28ms release delay. U102 CTS/CTR open: conditional 50us max fault detect.

REF2.5V / conditional safety2.34-2.53V. 20-30C static permission tested; thermal lag and transients NOT approved.

SYSTEM-P0: thermal R8 / JP1 removed; CORE JP1 is sole OPEN commissioning gate.

LAYOUT STUDY ONLY: two real offboard NTCs, no charge/fabrication approval.



SYSTEM-P0: core R6 and OPEN JP1 retained; NOFAB

Sheet: /thermal/
File: thermal.kicad_sch

Title: CE logic interlock candidate / NOFAB

Size: A2	Date: 2026-10-28	**Rev: NEW-REF-P0**
KiCad E.D.A. 9.0.2		Id: 3/3